# Demand and Welfare Analysis

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# Why Study Demand

- Starting point of investigating market structure and firm conduct
  - Market definition: cross-price elasticities of goods
  - Used to infer firm conduct: inputs of profit maximization
- Welfare analysis
  - Value of new/existing products
  - Policy evaluation: merger, tax, etc.
- Inform firm decisions (Marketing/Strategy)
  - Advertising, pricing, product introduction

- Recall the NEIO logic from the last lecture: marginal cost is *not* observed, it is inferred from a pricing FOC. That FOC contains $\partial q/\partial p$.
- **Demand estimates are the raw material for everything on the supply side.**

# Road-map from here

- **Toolkit**: the four objects of consumer theory, and the restrictions any demand system must respect.
- **Product space**: the Almost Ideal Demand System (AIDS), multi-stage budgeting, and the "Hausman IV".
  - Hausman, Leonard and Zona (1994): beer, mergers, conduct
- **Welfare**: $CV/EV$, why consumer surplus is only an approximation, and how to get the exact number (Hausman, 1981).
- **New goods**: virtual prices, Hausman (1996), Bresnahan's critique, and Chaudhuri, Goldberg and Jia (2006).

# Differentiated Product Demand

- Many of the basics were covered in lecture note section 1-3.
- We introduce two important approaches of estimating demand
  - Product space
  - Characteristic space
- Consumer heterogeneity is another important issue that Allan will cover
- Dynamics is left for later.

- The distinction is about **what indexes a product**: its own name (product space), or a vector of attributes (characteristic space).

# Product Space vs Characteristic Space

- Each has its own challenges and remedies
- Product space
  - Substitution pattern increases with number of products: "curse of dimensionality"
  - Remedy: impose additional structure of product nests or symmetry
  - Not well suited to consider hypothetical "new goods"
- Characteristic space
  - Product characteristics are either hard to observe or measure
  - Remedy: allow for unobserved product "quality" or characteristics is important, but could create additional econometric challenges
- We first cover the **Product Space** approach in this lecture.

# Toolkit

Denote prices $p$, income $w$, utility level $u$.

- **Marshallian (uncompensated) demand** $x(p, w) = \text{argmax}\ [u(x) : px \leq w]$ — what we *observe*.
- **Indirect utility** $v(p, w) = \max [u(x) : px \leq w]$.
- **Expenditure function** $e(p, u) = \min [px : u(x) \geq u]$ — the money metric.
- **Hicksian (compensated) demand** $h(p, u) = \text{argmin}\ [px : u(x) \geq u]$ — what we *want* for welfare.

Duality at the optimum:

$$e(p, v(p, w)) = w, \qquad v(p, e(p, u)) = u, \qquad x(p, e(p, u)) = h(p, u)$$

The welfare analysis is: **observe** $x$**, recover** $e$.

# Toolkit

- **Shephard's Lemma** — differentiate the expenditure function to get Hicksian demand:

$$\frac{\partial e(p,u)}{\partial p_i} = h_i(p,u), \qquad \text{equivalently} \qquad \frac{\partial \ln e(p,u)}{\partial \ln p_i} = \omega_i$$

  where $\omega_i \equiv p_i h_i / e$ is the **expenditure share**. *This is how AIDS is built.*

- **Roy's Identity** — differentiate the indirect utility function to get Marshallian demand:

$$x_i(p,w) = -\frac{\partial v(p,w)/\partial p_i}{\partial v(p,w)/\partial w}$$

  *This is how Hausman (1981) goes from an estimated demand curve to $v$.*

- **Slutsky equation** — links the two demands:

$$\frac{\partial x_i}{\partial p_j} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution}} - \underbrace{\frac{\partial x_i}{\partial w} x_j}_{\text{income}}$$

  In elasticity form: $\eta_{ij}^h = \eta_{ij} + \omega_j e_i$, where $e_i$ is the expenditure elasticity.

# Toolkit: What Any Demand System Must Satisfy

A share system $\omega_i(p, w)$ generated by utility maximization satisfies

- **Adding up**: $\sum_i \omega_i = 1$ for all $(p, w)$ (budget exhaustion).
- **Homogeneity**: $x(\lambda p, \lambda w) = x(p, w)$
- **Slutsky symmetry**: $\partial h_i / \partial p_j = \partial h_j / \partial p_i$. This is the *integrability* condition — lets us go back from demand to $e(p, u)$.
- **Negativity**: the Slutsky matrix is negative semi-definite; in particular $\partial h_i / \partial p_i \leq 0$.

Two practical implications used repeatedly below:

- Adding up makes one share equation **redundant**: estimate $J - 1$ of them and recover the last.
- Symmetry cuts free price coefficients from $J^2$ to $J(J + 1)/2$.

# The Curse of Dimensionality

- In product space, a flexible system with $J$ brands has $O(J^2)$ price coefficients.
  - Cereal: $J \approx 135$ brands $\Rightarrow$ over 18,000 price coefficients.
  - Beer: $J \approx 40$ brands $\Rightarrow$ still 1,600.
- Symmetry, homogeneity and adding-up help, but nowhere near enough.
- Worse: we need at least one **instrument per endogenous price**.
- **Two ways out**:
  1. Impose **separability** and estimate a small system at each of several levels (this lecture).
  2. Project products onto a low-dimensional **characteristic** vector (BLP; next lectures).
- Neither is free. Both buy tractability by restricting substitution patterns — know what you assumed, and test it if you can.

# Separability and Multi-Stage Budgeting

- Preferences are **weakly separable** in groups $G_1, \ldots, G_M$ if

$$u(x) = f\left(u_1(x_{G_1}), \ldots, u_M(x_{G_M})\right)$$

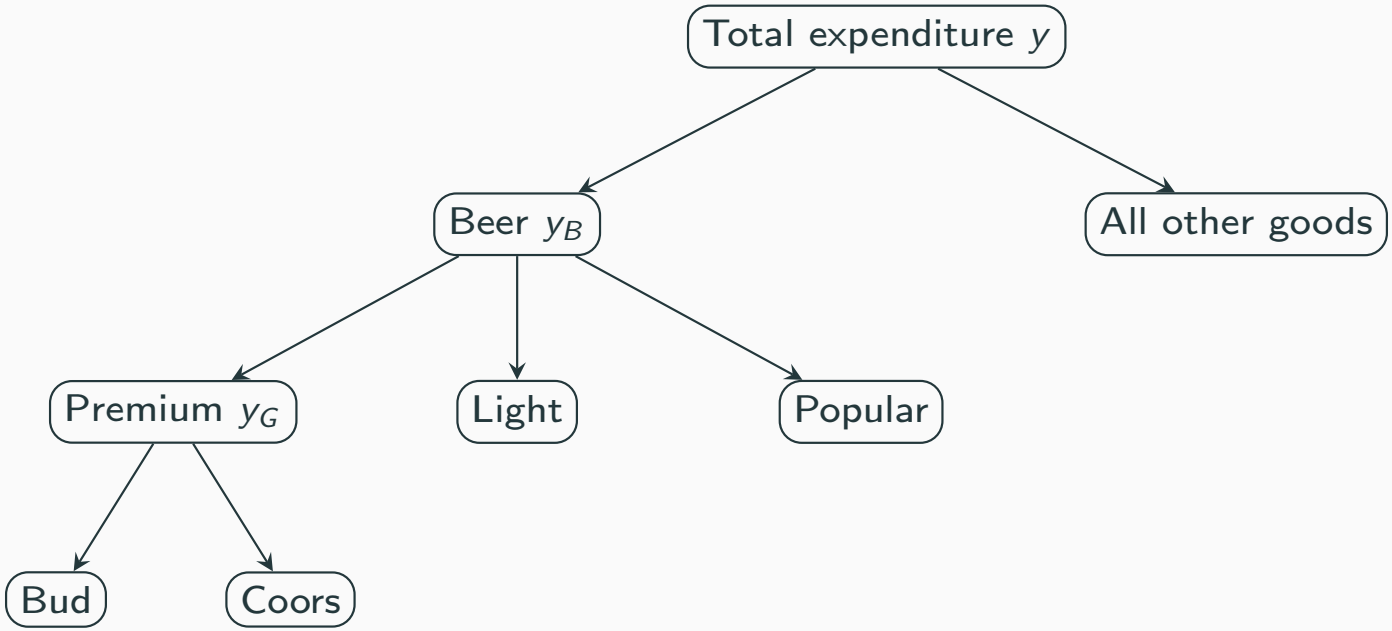
- Consequence: the *ranking* of bundles within a group does not depend on quantities outside the group. Conditional on group expenditure $y_m$, within-group demand depends only on within-group prices:

$$x_i = x_i(p_{G_m}, y_m), \qquad i \in G_m$$

- Gorman (1971): weak separability, plus conditions on the group price indices, lets the consumer's problem *decentralize* into stages — **multi-stage budgeting**.

- Cross-group substitution is no longer free. It is channelled entirely through group *expenditure* and group *price indices*. We will see exactly what that costs in HLZ.

# Three-Stage Budgeting: The Picture (HLZ Beer)




- **Top**: beer versus everything else. **Middle**: how beer expenditure splits across segments. **Bottom**: brand shares within a segment.
- Estimated *bottom up*: the bottom level delivers the price indices the middle level needs, and so on.
- Segments come from **industry/marketing practice**, not from the econometrics — but they are testable (we return to this).

# The AIDS Demand

- Starting from the **expenditure function** $e(p, u)$ (Deaton and Muellbauer, 1980)

$$\ln(e(p,u)) = \alpha_0 + \sum_k \alpha_k \ln p_k + \frac{1}{2}\sum_k \sum_j \gamma^*_{kj} \ln p_k \ln p_j + u\beta_0 \Pi_k p_k^{\beta_k}$$

- This is the **PIGLOG** class, $\ln e(p,u) = (1-u)\ln a(p) + u \ln b(p)$, $u \in [0,1]$: the cost of bliss. The first three terms are $\ln a(p)$, a **translog** aggregator — a second-order (flexible) approximation to *any* price index.
- Directly use the Shephard's Lemma, we have the Hicksian demand system

$$\omega_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln [w/P], \forall i$$

$$\ln P = \alpha_0 + \sum_k \alpha_k \ln p_k + \frac{1}{2}\sum_j \sum_k \gamma_{kj} \ln p_k \ln p_j$$

  where $\gamma_{ij} = \frac{1}{2}(\gamma^*_{ij} + \gamma^*_{ji})$

- Once demand system is estimated, the functional form of $e(p,u)$ is known. *CV* (compensating variation) can be easily evaluated.

# The AIDS Demand

Detailed derivation for the $\ln(w/P)$ term.

1. Shephard in log form gives the **Hicksian** share directly:

$$\omega_i = \frac{\partial \ln e(p,u)}{\partial \ln p_i} = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \, u\beta_0 \prod_k p_k^{\beta_k}$$

2. This still contains the unobservable $u$. Evaluate at the optimum, where $e(p,u) = w$, and invert for the last term:

$$u\beta_0 \prod_k p_k^{\beta_k} = \ln w - \ln P(p)$$

3. Substitute back to get the **Marshallian** share equation, in observables only:

$$\omega_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln [w/P]$$

Reading the parameters: $\gamma_{ij}$ is the share response to a relative price change; $\beta_i > 0$ means good $i$ is a **luxury** within the group (its share rises with real expenditure), $\beta_i < 0$ a necessity.

# AIDS: Restrictions from Theory

The four properties above translate into *linear* restrictions on parameters — a large part of why AIDS is convenient.

- **Adding up** ($\sum_i \omega_i = 1$):

$$\sum_i \alpha_i = 1, \qquad \sum_i \gamma_{ij} = 0 \ \ \forall j, \qquad \sum_i \beta_i = 0$$

- **Homogeneity of degree zero** in $(p, w)$:

$$\sum_j \gamma_{ij} = 0 \ \ \forall i$$

- **Slutsky symmetry**:

$$\gamma_{ij} = \gamma_{ji}$$

- **Negativity** is not a linear restriction; check it after estimation, typically at the sample mean.

# AIDS: Why It Became the Workhorse

- **Flexible**: an arbitrary first-order approximation to any demand system — price elasticities are unrestricted at the point of approximation. No IIA.
- **Consistent with theory**: derived from an expenditure function, so integrability holds *by construction*. We get $e(p, u)$ for free, hence exact welfare.
- **Aggregates exactly** over consumers without requiring parallel linear Engel curves (the PIGLOG), so market-level share data is useful.
- **Linear in parameters** if $\ln P$ is replaced by the **Stone index**

  $$\ln \tilde{P} = \sum_k \omega_k \ln p_k$$

  so the system can be estimated by SUR rather than nonlinear GMM ("LA/AIDS").

- Caveats on Stone: (i) $\omega_k$ on the right-hand side is itself endogenous — often replaced by lagged or mean shares; (ii) the approximation is not invariant to units of measurement. Alston, Foster and Green (1994) find by Monte Carlo that it is nonetheless a good approximation in practice.

# Demand Elasticity

- Hausman et al (1994) used Stone approximation

$$\log P = \sum_j \omega_j \log p_j$$

- We first show that $\frac{d\omega_i}{d \log p_j}$ is connected to demand elasticities $\frac{d \log q_i}{d \log p_j}$.
  If $i = j$, i.e., we are interested in own elasticity, then

$$\frac{d\omega_i}{d \log p_j} = \frac{d(p_i q_i)}{d \log p_i} \cdot \frac{1}{y_G} = [p_i q_i + \frac{dq_i}{d \log p_i} p_i]/y_G \equiv \omega_i[1 + \eta_{ii}]$$

  where $\eta_{ii} = \frac{d \log q_i}{d \log p_i}$.

- If $i \neq j$, i.e., we are interested in cross elasticity, then

$$\frac{d\omega_i}{d \log p_j} = \frac{d(p_i q_i)}{d \log p_j} \cdot \frac{1}{y_G} = [0 + \frac{dq_i}{d \log p_j} p_i]/y_G \equiv \omega_i \eta_{ij}$$

  where $\eta_{ij} = \frac{d \log q_i}{d \log p_j}$.

# Demand Elasticity

- Making use of the AIDS demand system, we know that

$$\frac{d\omega_i}{d\log p_j} = \gamma_{ij} - \beta_i \frac{d\log P}{d\log p_j}$$

- As a result, we have the demand elasticity (given $y_G$)

$$\eta_{ii} = -1 + [\gamma_{ii} - \beta_i \frac{d\log P}{d\log p_i}]/\omega_i$$

$$\eta_{ij} = [\gamma_{ij} - \beta_i \frac{d\log P}{d\log p_j}]/\omega_i$$

- The calculation of $\frac{d\log P}{d\log p_j}$ can either use AIDS exact price aggregation

$$\frac{d\log P}{d\log p_j} = \alpha_j + \sum_k \gamma_{kj} \log p_k$$

or use Stone approximation formula $\frac{d\log P}{d\log p_j} = \omega_j$

# Hausman, Leonard, Zona (1994): Overview

- Studies American domestic beer industry and conduct merger analysis.
- Data: weekly store-level scanner (IRI/Nielsen-type) data, by brand and city — a **panel** across cities $n$ and weeks $t$.

- Two contributions worth separating:
  1. **Demand**: a flexible, theory-consistent brand-level system that does *not* impose IIA, made tractable by multi-stage budgeting.
  2. **Applications**: (i) merger simulation; (ii) inference on competitive interaction. Note these use the demand system as a *primitive* — no conduct assumption is needed to estimate demand itself.

- The paper is also the standard citation for the **"Hausman instrument"**, which we will look at carefully — and which Bresnahan attacks.

# Hausman, Leonard, Zona (1994)

- Studies American domestic beer industry and conduct merger analysis.
- Embed the AIDS demand system into a three-stage budgeting process
  - Bottom level: market share of brand $i$ in city $n$ at time $t$

$$\omega_{int} = \alpha_{in} + \sum_j \gamma_{ij} \log(p_{jnt}) + \beta_i \log\left[\frac{y_{Gnt}}{P_{nt}}\right] + \epsilon_{int}, \forall i, n, t$$

    where $y_{Gnt}$ is segment-beer expenditure
  - Middle level: segment demand is log-log

$$\log q_{mnt} = \alpha_{mn} + \beta_m \log y_{Bnt} + \sum_k \delta_k \log(\pi_{knt}) + \epsilon_{mnt}$$

    where $y_{Bnt}$ is beer expenditure and $\pi_{knt}$ is segment price index (i.e. $P_{nt}$ at the bottom level).
  - Upper level (i.e. buy beer overall) is a similar log-log formulation.
- Segments: **premium**, **popular** (non-premium), **light**, **imported**.

# HLZ: What Separability Buys, and What It Costs

- **Within** a segment, substitution is completely unrestricted: the $\gamma_{ij}$ matrix is free (up to symmetry and adding up). This is the whole point — Bud and Miller are allowed to be much closer than Bud and Genesee.

- **Across** segments, weak separability implies the compensated cross-price elasticity between brand $1 \in G$ and brand $2 \in H$ takes the form

$$\delta_{12} = \lambda_{GH} \frac{\partial \log q_1}{\partial \log y_G} \frac{\partial \log q_2}{\partial \log y_H}$$

  where $\lambda_{GH}$ depends only on the *segments*, not on the brands.

- Cross-segment substitution is a **rank-one** object. Two premium brands can substitute differently for a given light brand *only* through their segment-expenditure elasticities.

- So the segment structure is not innocuous.

# HLZ: Testing the Segment Structure

- The restriction above gives a clean specification test. Take the **middle-level** equation for premium beer and add the *individual brand prices* from the light segment as extra regressors.
- Under weak separability those individual prices should not matter — only the light-segment *price index* should enter, and only through the segment allocation.
- HLZ's finding: none of Genesee Light, Coors Light, Old Milwaukee Light or Molson Light is significant; Miller Lite is marginally significant. The estimated premium-segment elasticities do not change.
- Conclusion: the segment structure is "adequate for estimation", with the caveat that further work on segment definition is warranted.
- Note the general lesson, which recurs in the BLP literature: *a nesting structure is an assumption with testable implications*.

# HLZ: From Conditional to Overall Elasticities

The bottom level gives elasticities *conditional* on segment expenditure. The number a merger needs is the **overall** elasticity, which lets segment and total beer expenditure adjust. Chain rule, for $i, j$ in segment $m$:

$$\underbrace{\eta_{ij}}_{\text{overall}} = \underbrace{\eta^{c}_{ij}}_{\text{bottom}} + \underbrace{e_i}_{\substack{\text{expenditure}\\ \text{elasticity}}} \times \underbrace{\frac{\partial \ln y_m}{\partial \ln p_j}}_{\text{middle \& top}}$$

with the three ingredients

- $e_i = 1 + \beta_i/\omega_i$ (bottom level)
- $\partial \ln \pi_m / \partial \ln p_j = \omega_j$ (Stone index of the segment)
- $\dfrac{\partial \ln y_m}{\partial \ln p_j} = \big(1 + \delta_{mm}\big)\omega_j \;+\; \beta_m\big(1 + \delta_B\big)\, W_m \omega_j$

where $\delta_{mm}$ is the own-price-index elasticity of segment quantity, $\beta_m$ the segment expenditure elasticity, $\delta_B$ the top-level own-price elasticity of beer, and $W_m$ segment $m$'s share of beer expenditure. Since $\delta_{mm} < -1$ empirically (Table 1: $-2.67$ for premium), the second term is **negative**: allowing segment expenditure to fall makes overall demand *more* elastic than conditional demand. Compare Budweiser: $-3.53$ conditional (Table 2) vs. $-4.20$ overall (Table 5).

# Identification and Estimation Details

- Endogeneity of price: multiple reasons that price $p_{jnt}$ could be correlated with unobservable $\epsilon_{int}$.
- Note that the multiple market/periods setting here allow for $\alpha_{in}$, which controls for many time-invariant product characteristics. However, $\epsilon_{int}$ could still include many promotion activities (and increase marginal cost/price of selling a product).
- Hausman et al proposes an empirical specification

$$\log p_{jnt} = \delta_j \log c_{jt} + \alpha_{jn} + w_{jnt}$$

where $c_{jt}$ contains production and national advertising cost. $\alpha_{jn}$ captures city-specific logistic cost/wage differentials. $w_{jnt}$ is city-specific promotion cost (source of endogeneity).

- KEY assumption: $w_{jnt}$ is independent across cities. In other words, $w_{jnt} \perp \epsilon_{jn't}, \forall n' \neq n$. As a result, we can use prices (or average prices $c_{jt}$) of other cities as IVs, once controlling for brand-market fixed effects. This is called "Hausman IV".

# The Hausman IV

Decompose the price of brand $j$ in city $n$ at time $t$ into three orthogonal pieces:

$$\log p_{jnt} = \underbrace{\delta_j \log c_{jt}}_{\text{national cost, varies over } t} + \underbrace{\alpha_{jn}}_{\text{city}\times\text{brand, fixed}} + \underbrace{w_{jnt}}_{\text{local, transitory}}$$

- **Need**: a shifter of $p_{jnt}$ that is orthogonal to the local demand shock $\epsilon_{jnt}$.
- **The claim**: $p_{jn't}$ for $n' \neq n$ contains $\delta_j \log c_{jt}$ — a common, time-varying *cost* component — but its own local shock $w_{jn't}$ is independent of $\epsilon_{jnt}$.
- So, **after brand–city fixed effects**, prices in other cities are valid instruments. This is Hausman and Taylor (1981) variance-components identification applied to a panel of markets.

- In practice, either use $\overline{p}_{j,-n,t}$ (leave-one-out mean across cities) or the full set of other-city prices.
- Attractive because **excluded cost shifters at the brand level are hard to find**: brands within a segment share inputs, technology and wages.

# When Does the Hausman IV Fail?

Bresnahan's (1996) comment on Hausman is the standard critique:

- The identifying assumption requires $\epsilon_{jnt}$ to have **no national component**. That rules out national advertising campaigns, national fads, and slow national acceptance of a brand.
- If there *is* a national demand shock and supply slopes up (which the Bertrand pricing equations themselves imply), then $w_{jnt}$ is correlated across cities and the instrument is invalid.

- **Direction of the bias**: in a linear demand / Bertrand example, Bresnahan shows the own-price coefficient is biased *toward zero* — demand looks **too steep**, i.e., too inelastic. I.e., ordinary simultaneity bias, transmitted through the other cities.
- That is the direction that inflates estimated market power and consumer surplus from new goods.

- Defenses: control directly for national advertising; exploit market-specific timing; or switch to a different exclusion restriction (BLP-type characteristic instruments).

# HLZ: Estimation Mechanics

- Estimate **bottom up**: the bottom level delivers $\hat{\pi}_{mnt}$, the segment price indices needed at the middle level; the middle level delivers the beer price index for the top.
- The bottom-level share system is estimated as a **system** (SUR) to exploit cross-equation correlation in $\epsilon_{int}$ and to impose symmetry *across* equations.
- Adding up implies $\sum_i \epsilon_{int} = 0$: the residual covariance matrix is singular. **Drop one equation**; the estimates are invariant to which one.
- Brand–city fixed effects $\alpha_{in}$ absorb time-invariant tastes and distribution costs; a time trend and the number of stores absorb coverage changes.

# Results: Middle Level (Segment Demands)

TABLE 1

***Beer Segment Conditional Demand Equations.***

| | Premium | Popular | Light |
|---|---|---|---|
| Constant | 0.501 | −4.021 | −1.183 |
| | (0.283) | (0.560) | (0.377) |
| log (Beer Exp) | 0.978 | 0.943 | 1.067 |
| | (0.011) | (0.022) | (0.015) |
| log ($P_{PREMIUM}$) | −2.671 | 2.704 | 0.424 |
| | (0.123) | (0.244) | (0.166) |
| log ($P_{POPULAR}$) | 0.510 | −2.707 | 0.747 |
| | (0.097) | (0.193) | (0.127) |
| log ($P_{LIGHT}$) | 0.701 | 0.518 | −2.424 |
| | (0.070) | (0.140) | (0.092) |
| Time | −0.001 | −0.000 | 0.002 |
| | (0.000) | (0.001) | (0.000) |
| log (# of Stores) | −0.035 | 0.253 | −0.176 |
| | (0.016) | (0.034) | (0.023) |

Number of Observations = 101.

- Segment quantity is very elastic in its own price index ($-2.67$ premium, $-2.71$ popular, $-2.42$ light) — this is what drives the wedge between conditional and overall brand elasticities.
- Beer expenditure elasticities $\approx 1$: segment shares are roughly stable in beer expenditure.

# Results: Bottom Level (Brand Shares, Premium)

TABLE 2

***Brand Share Equations: Premium.***

| | 1 Budweiser | 2 Molson | 3 Labatts | 4 Miller | 5 Coors |
|---|---|---|---|---|---|
| Constant | 0.393 | 0.377 | 0.230 | −0.104 | – |
| | (0.062) | (0.078) | (0.056) | (0.031) | – |
| Time | 0.001 | −0.000 | 0.001 | 0.000 | – |
| | (0.000) | (0.000) | (0.000) | (0.000) | – |
| log (Y/P) | −0.004 | −0.011 | −0.006 | 0.017 | – |
| | (0.006) | (0.007) | (0.005) | (0.003) | – |
| log ($P_{Budweiser}$) | −0.936 | 0.372 | 0.243 | 0.150 | – |
| | (0.041) | (0.231) | (0.034) | (0.018) | – |
| log ($P_{Molson}$) | 0.372 | −0.804 | 0.183 | 0.130 | – |
| | (0.231) | (0.031) | (0.022) | (0.012) | – |
| log ($P_{Labatts}$) | 0.243 | 0.183 | −0.588 | 0.028 | – |
| | (0.034) | (0.022) | (0.044) | (0.019) | – |
| log ($P_{Miller}$) | 0.150 | 0.130 | 0.028 | −0.377 | – |
| | (0.018) | (0.012) | (0.019) | (0.017) | – |
| log (# of Stores) | −0.010 | 0.005 | −0.036 | 0.022 | – |
| | (0.009) | (0.012) | (0.008) | (0.005) | – |
| Conditional Own Price Elasticity | −3.527 | −5.049 | −4.277 | −4.201 | −4.641 |
| | (0.113) | (0.152) | (0.245) | (0.147) | (0.203) |

- Notice that market shares add up to one, needs to drop one brand (Coors here — the "–" column).
- Symmetry is visible in the table: $\gamma_{\text{Bud,Molson}} = \gamma_{\text{Molson,Bud}} = 0.372$.
- Small $\beta_i$ (on $\log(Y/P)$): brand shares barely respond to segment expenditure, so $e_i \approx 1$.

# Results: Overall Elasticities (Light Segment)

***Light Segment Own and Cross Elasticities.***

| | Genesee Light | Coors Light | Old Milwaukee Light | Lite | Molson Light |
|---|---|---|---|---|---|
| Genesee Light . . . . . | −3.763 | 0.464 | 0.397 | 0.254 | 0.201 |
| | (0.072) | (0.060) | (0.039) | (0.043) | (0.037) |
| Coors Light . . . . . | 0.569 | −4.598 | 0.407 | 0.452 | 0.482 |
| | (0.085) | (0.115) | (0.058) | (0.075) | (0.061) |
| Old Milwaukee Light . . | 1.233 | 0.956 | −6.097 | 0.841 | 0.565 |
| | (0.121) | (0.132) | (0.140) | (0.112) | (0.087) |
| Lite . . . . . | 0.509 | 0.737 | 0.587 | −5.039 | 0.577 |
| | (0.095) | (0.122) | (0.079) | (0.141) | (0.083) |
| Molson Light . . . . . | 0.683 | 1.213 | 0.611 | 0.893 | −5.841 |
| | (0.124) | (0.149) | (0.093) | (0.125) | (0.148) |

- Own elasticities $-3.8$ to $-6.1$: implied Lerner indices of roughly 0.16–0.27 under single-product Bertrand.
- Cross elasticities are **asymmetric** and vary a lot (0.20 to 1.23) — exactly what IIA would rule out. This is the payoff of the flexible product-space specification.

# Application: Merger Analysis

- Traditional approach: define the market, compute HHI, apply the Merger Guidelines thresholds. HLZ instead go **directly to prices**.
- Take Nash–Bertrand as the pricing model. A single-product firm sets

$$\frac{p_1 - mc_1}{p_1} = -\frac{1}{e_{11}}$$

- After merging firms producing $j = 1, \ldots, m$, the combined firm internalizes the diverted demand:

$$s_j + \sum_{k=1}^{m} \left[ \frac{p_k - mc_k}{p_k} s_k \right] e_{kj} = 0, \qquad j = 1, \ldots, m$$

- Stack this as a linear system in the (share-weighted) markups $w$:

$$s + E'w = 0 \qquad \Longrightarrow \qquad w = -(E')^{-1}s$$

  where $s$ is the vector of revenue shares and $E$ the matrix of cross-price elasticities.

- **Nothing here required estimating the supply side**: given Bertrand and the demand system, the post-merger markups are a matrix inversion.

# Merger Analysis: Prices and Efficiencies

- Write the markup as $\theta_j = (p_j - mc_j)/p_j$, and $\theta_j^M$ for its post-merger value. Then

$$\frac{p_j^M}{p_j} = \frac{mc_j^M / mc_j}{\left(\frac{p_j}{mc_j}\right)(1 - \theta_j^M)} \qquad \text{and} \qquad \frac{p_j}{mc_j} = \frac{e_{jj}}{1 + e_{jj}}$$

- The numerator is the **efficiency** channel: any fall in marginal cost passes through proportionally. The denominator contains $\theta_j^M$, the **market power** channel.
- So a merger simulation is a **horse race**: cost synergies against internalized diversion. Both sides are quantified by the same demand estimates — no separate cost data needed.

- Caveat, the same one Bresnahan will raise for new goods: a proper counterfactual re-solves the **full** Bertrand system, **letting non-merging firms re-optimize too**.

# Welfare: What We Actually Want

- Elasticities answer "what happens to quantities". Policy questions ask "by how much are consumers better or worse off, in dollars".

- Three questions we will need to answer:
  1. A **price** changes (merger, tax, tariff, patent enforcement).
  2. A **product appears** (Apple-Cinnamon Cheerios, the minivan).
  3. A **product disappears** (generic drugs withdrawn under TRIPS).

- All three are the same computation: an integral under a **Hicksian** demand curve, i.e., a difference of expenditure functions. Case 2 and 3 just require evaluating it at a price we never observe.

- So the practical question is: how do we get $e(p, u)$ when all we see is $x(p, w)$?

# Welfare: Review of the Basics

- Traditional methods have relied on **Compensating Variation (CV)** and **Equivalent Variation (EV)**

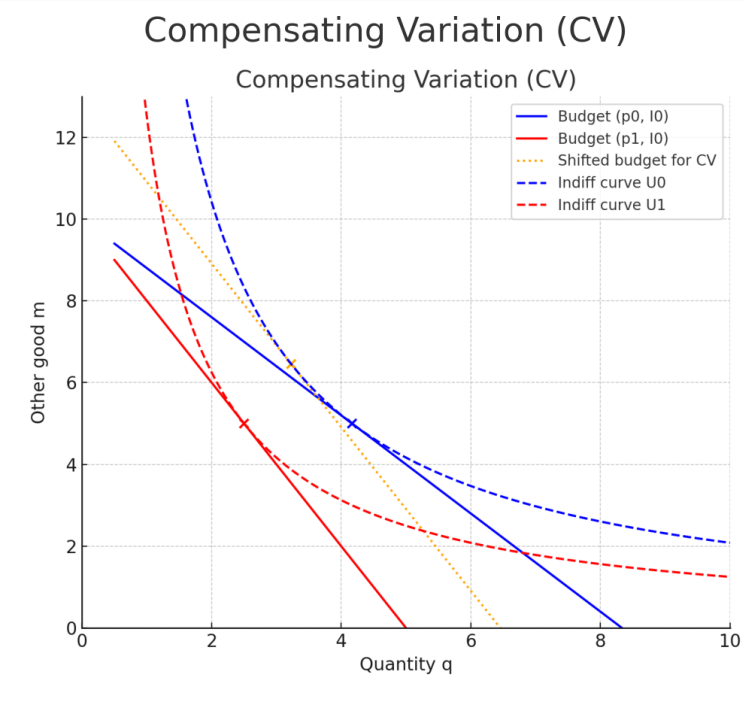
- Consider a change in prices from $p$ to $p'$ (with income fixed at $w$)
  - Utility goes from $u = v(p, w)$ to $u' = v(p', w)$
  - The change in income that would compensate the change in prices (previous $u$, new price $p'$)

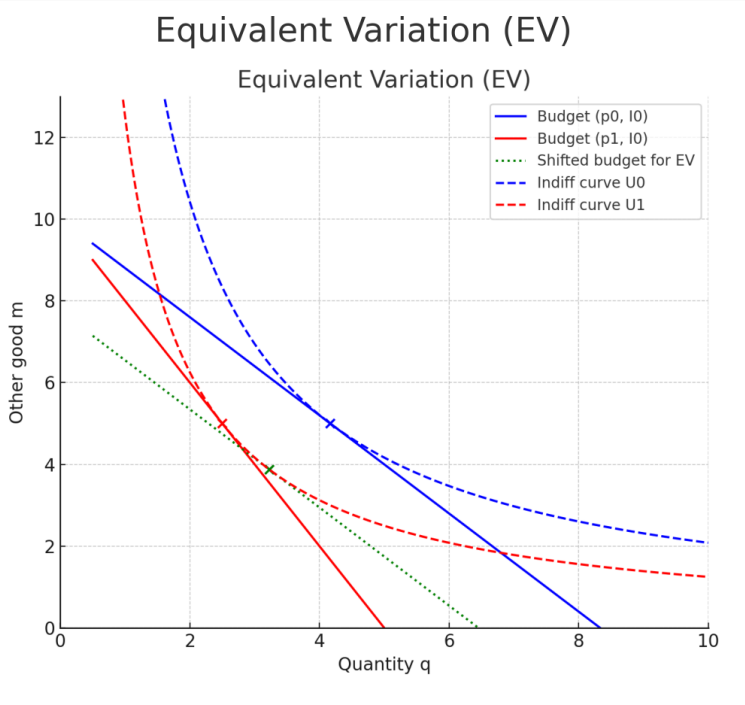
$$CV = e(p, u) - e(p', u) = w - e(p', u)$$
$$\mathrm{v}(\mathrm{p}', \mathrm{w} + \mathrm{CV}) = \mathrm{v}(\mathrm{p}, \mathrm{w})$$

  - The change in income that would be equivalent to the change in prices (new utility $u'$, previous price $p$)

$$EV = e(p, u') - e(p', u') = e(p, u') - w$$

# Welfare: Review of the Basics



# Welfare: Review of the Basics

- With the definitions above, both $CV$ and $EV$ are **positive for a price fall** and **negative for a price rise**. They are signed *gains*.
- Most applied papers instead report a **loss**, i.e., the positive number

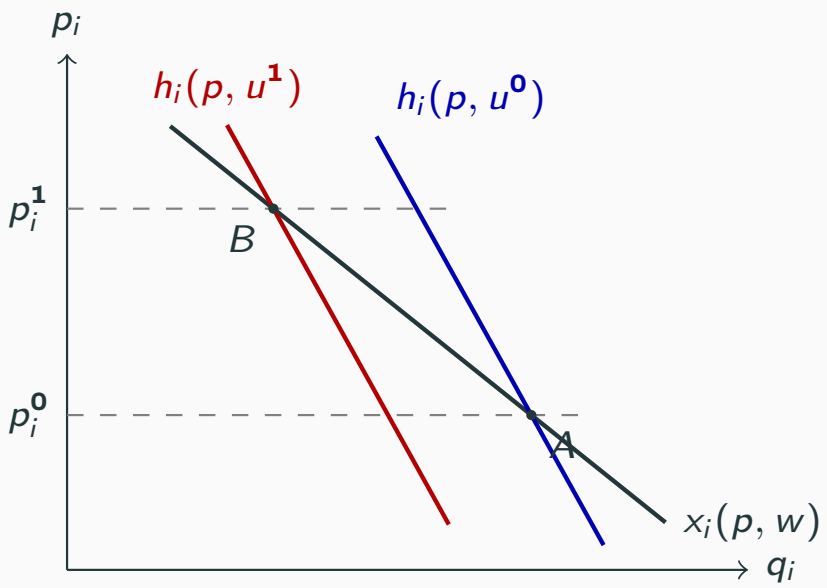
  $$L_{CV} = e(p', u) - w, \qquad L_{EV} = w - e(p, u')$$

  for a price increase $p \to p'$. Chaudhuri–Goldberg–Jia's "compensating variation" is $L_{CV}$ in this sense.
- Below we use the **loss** convention when ranking $CV$, $CS$ and $EV$, i.e., all three are positive magnitudes of the welfare change.

# Welfare: Review of the Basics

- For price changes in only one good $i$, CV and EV can be computed based on Hicksian Demand curve
  - $CV = \int_{p_i'}^{p_i} \frac{\partial e(p,u)}{\partial p_i} dp_i = \int_{p_i'}^{p_i} h_i(p,u) dp_i$
  - $EV = \int_{p_i'}^{p_i} \frac{\partial e(p,u')}{\partial p_i} dp_i = \int_{p_i'}^{p_i} h_i(p,u') dp_i$
- Can similarly define the consumer surplus (CS) with Marshallian Demand curve
  - $CS = \int_{p_i'}^{p_i} x_i(p,w) dp_i$
  - At end points $x_i(p',w) = h_i(p',u')$ and $x_i(p,w) = h_i(p,u)$
- Can we graphically show (for a normal good), EV $<$ CS $<$ CV? When does $EV = CS = CV$?

# Why $EV < CS < CV$ (Normal Good)




- Both Hicksian curves are **steeper** than the Marshallian (Slutsky, normal good). $h(\cdot, u^0)$ passes through $A$, $h(\cdot, u^1)$ through $B$.
- Between $p_i^0$ and $p_i^1$: $h(\cdot, u^1) \leq x \leq h(\cdot, u^0)$, so the areas to the left rank $EV < CS < CV$ (losses).
- Equality iff $\partial x_i / \partial w = 0$ (quasi-linear; $\beta_i = 0$ in AIDS) — then Marshallian = Hicksian.

# Welfare Analysis: General Demand

- Evaluating $CV$ or $EV$ with an observed market demand curve is challenging. $\rightarrow$ Needs unobserved Hicksian demand. Hausman (1981) provides a solution:
  - Define the indirect utility function $v(p, w) \equiv max\,[u(x) : px \leq w]$
  - Roy's identity $x_i(p, w) = -\frac{\partial v(p,w)/\partial p_i}{\partial v(p,w)/\partial w}$. Solve differential equation to retrieve $v(p, w)$
  - Use $v(p, e(p, u)) = u$ to invert for expenditure function $e(p, u)$
  - Then use either $e(p, u)$ (or $h_i(p, u)$) to get $CV$ or $EV$

- The procedure can accommodate non-parametric demand curve, but often cumbersome to implement in practice (think of large number of products)[1].

- **Demand system that specify parametric form of $e(p, u)$ or $v(p, w)$ are often easier to work with**.

[1] One exception is the CES demand system.

# Welfare with AIDS: Why It Is Easy

- With AIDS we never have to solve a differential equation, because we *started* from $e(p, u)$. Estimation delivers $\hat{\alpha}, \hat{\beta}, \hat{\gamma}$; the expenditure function is then known in closed form.

- For a price change $p^0 \to p^1$ with $u^0$ the pre-change utility,

  $$L_{CV} = e(p^1, u^0) - e(p^0, u^0) = e(p^1, u^0) - w$$

  and $u^0$ is recovered by inverting $e(p^0, u^0) = w$ — one line of algebra in the PIGLOG form.

- Because $\ln e$ is additively separable into $\ln a(p)$ and $u \cdot b(p)$, the computation is a pair of index-number evaluations.

- **The benefit of product-space systems in welfare work**: the money metric is a by-product of estimation, and it is exact.
- The cost, as always, is the dimensionality and the inability to price a good that has never existed — unless we use a virtual price.

# New Goods: The Problem

- The CPI links a new good in only after it has been on the shelf for a while. Implicitly, the new good either does not exist or is a *perfect substitute* for what is already there. Both understate its value.

- Bresnahan puts the issue precisely: assuming perfect substitutability is assuming the new good's demand curve is **flat**. The worse the substitutability, the steeper the curve, and the bigger the consumer surplus triangle that gets thrown away.

- Two economically distinct questions get conflated:
  1. How much do consumers value the *new variety* per se?
  2. How does the introduction change *prices* of everything else, through competition?

- The classical theory answers (1) under perfect competition. Answering (2) requires a supply side — which is where Bresnahan and, later, CGJ come in.

# Virtual (Reservation) Prices

- Hicks (1940) and Rothbarth (1941): treat a good that is unavailable as a good that is **rationed to zero**. A rationed consumer behaves as if facing a higher, hypothetical price — the **virtual price** $p_n^V$ — at which she would *voluntarily* choose zero.

- Formally, $p_n^V$ solves

$$0 = x_n\left(p_1, \ldots, p_{n-1}, p_n^V, w\right)$$

- Once we have $p_n^V$, valuing the new good is just a price change from $p_n^V$ to the observed $p_n^0$:

$$CV = e\left(p_n^V, p_{-n}, u^0\right) - e\left(p_n^0, p_{-n}, u^0\right)$$

- **Nothing new is needed** — the machinery is exactly the price-change machinery. The hard part is that $p_n^V$ lies outside the support of the data.

# The AIDS Demand: New Variety

- What about a new variety is introduced (i.e. Hausman 1994), or excluded (Chaudhuri, Goldberg, Jia 2006)?

- Hausman (1994) uses the earlier "rationing" literature and define the "virtual price" of the new variety $n$ as

$$0 = x_n = g(p_1, ..., p_{n-1}, p_n^*, w)$$

- For an AIDS demand system, the virtual price $p_n^V$ is defined by

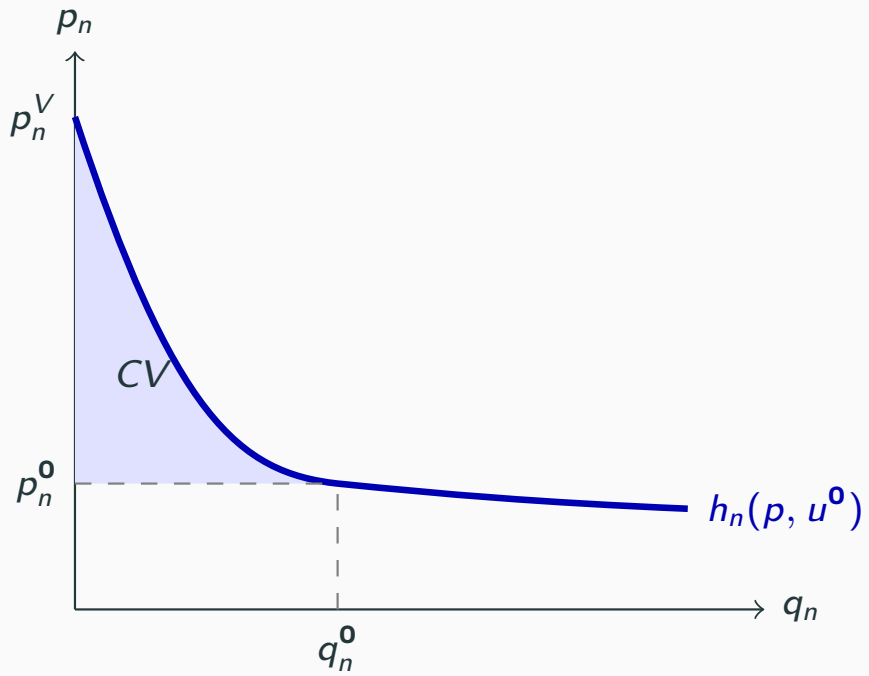
$$0 = \alpha_n + \gamma_{nn} \ln p_n^V + \sum_{j \neq n} \gamma_{nj} \ln p_j + \beta_i \ln [w/P]$$

  In practice, we often have data from multiple markets (or time period), so Hausman (1994) traces out an "average" demand curve of product $n$.

- Then compute the CV from virtual price $P_n^V$ to the observed price.

- Note the AIDS virtual price is available **in closed form**: set the share to zero and solve a linear equation in $\ln p_n^V$. That is a practical advantage.

# Virtual Price: The Picture




- The shaded area is the gain from the introduction. It is governed entirely by **how steep** the demand curve is near the axis.
- But that region is **never observed**: prices as high as $p_n^V$ do not occur in the data. The number is a functional-form extrapolation.
- Hausman's own illustration: fitting a CES instead of AIDS makes the estimate about **three times larger** (and $p_n^V$ infinite).

# Hausman (1996): Application

Valuation of New Goods under Perfect and Imperfect Competition. Case study: **Apple-Cinnamon Cheerios**, introduced by General Mills in 1989.

- Why this brand? It is deliberately a "me too" product — General Mills already sold Cheerios and Honey-Nut Cheerios. If *this* product has large consumer surplus, incremental variety in branded consumer goods matters generally.

- Same three-level structure as HLZ: top (cereal vs. other goods), middle (adult / child / family segments), bottom (brands within segment). AIDS at the bottom, Hausman IV across cities.

- Scale: RTE cereal saw roughly 190 new brands introduced into a pool of about 160 between 1980–92; over 95 were discontinued. Only 2 of the 190 reached a 1% share. Yet about 25% of consumption comes from brands under ten years old.

- Apple-Cinnamon Cheerios: 6.19% of the family segment, 1.6% of RTE cereal overall. Family segment is 26.4% of the market.

Hausman (1994) Figure 5.1.

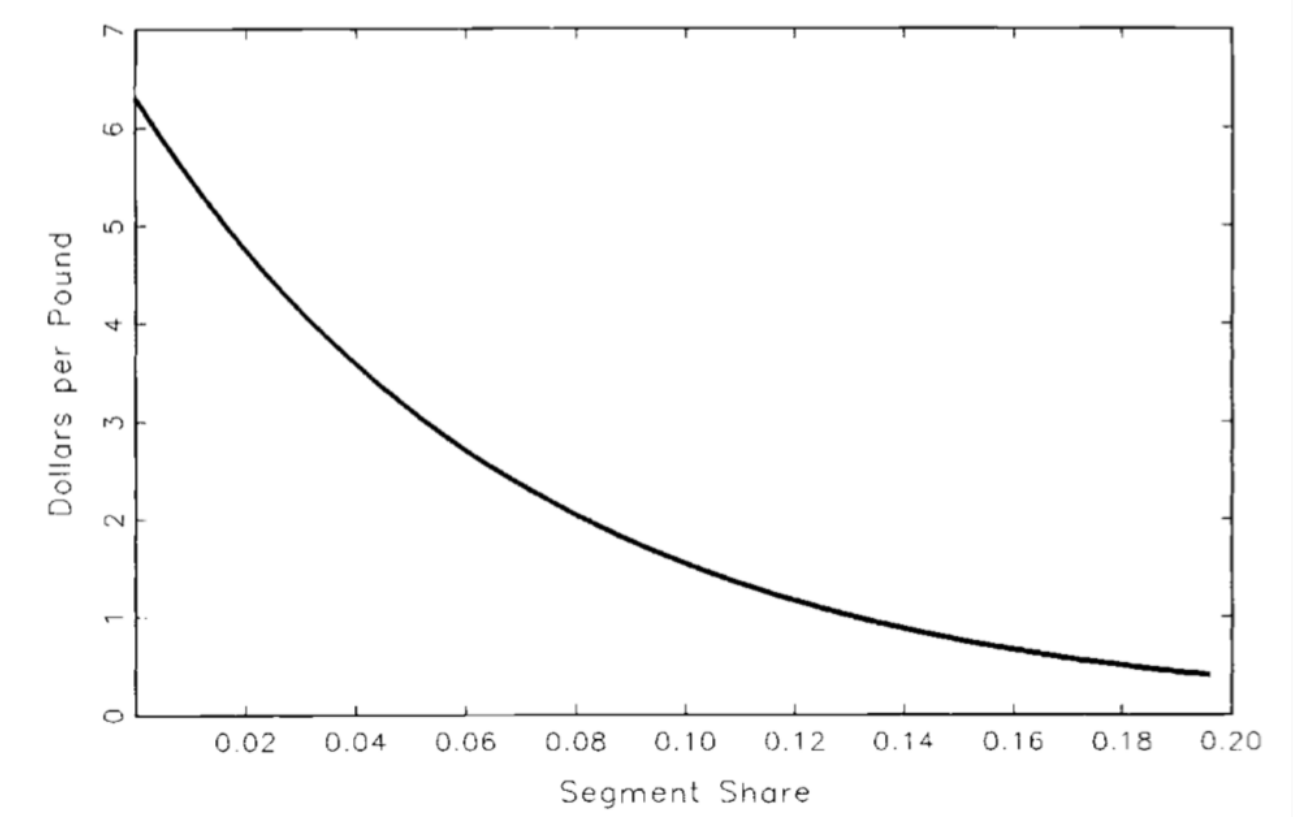



**Fig. 5.1 Almost-ideal demand curve for Apple-Cinnamon Cheerios (method 1)**

- Found an estimate of $32,268$ per city, weekly average, for Apple-Cinnamon Cheerios. Overall, $66.8 - 78.1$ million per year.

# Hausman (1996): The Magnitudes

- Virtual price $\hat{p}^V = \$7.14$ (asymptotic s.e. \$1.33) against an average actual price of \$3.48 — roughly **twice** the market price. Implied own-price elasticity about $-2.2$.
- Exact consumer surplus: \$32,268 per city per week (s.e. \$3,384), i.e., \$78.1m per year for the US, or \$0.31 per person per year.
- A convexity-based **lower bound** (tangent to the demand curve at the observed price) still puts the reservation price at \$5.55–\$5.94, about 65% above the actual price.
- Adjusting for imperfect competition (General Mills re-prices Cheerios and Honey-Nut Cheerios upward after the introduction): the net gain falls to \$27,395, i.e., 85.6% of the Hicksian number, or \$66.8m per year.
- Aggregating up: the cereal CPI may be overstated by about 25% (perfect competition) or 20% (imperfect competition) because new brands are ignored.
- Provocative claim, and a heavily contested one. Two lines of attack follow.

# The AIDS Demand: New Variety

- Bresnahan commented on Hausman (1994)'s magnitude based on
  - Under-estimate the cross-substitution due to the "Hausman IV" (i.e. no national advertising)
  - Lack of oligopolistic interaction and supply side adjustment, i.e., other firms/brands' prices were held as *fixed*
- Chaudhuri Goldberg and Jia (2006) addressed the second by re-calculating optimal prices of all the other products in market and the virtual price equation *jointly*.
- We take the two critiques in turn.

# Bresnahan's Critique I: The Instrument

- Hausman's estimate is large *because* the estimated demand curve for Apple-Cinnamon Cheerios is steep — i.e., poor substitutability with existing family-segment cereals.

- Bresnahan concedes the *specification* is not the culprit: within-segment substitution is left unrestricted, and the segment structure is credible (marketing departments have done that work).

- The suspect is the **identification**. The Hausman IV requires no national brand-level demand shocks. But national advertising campaigns, fads, and slow brand acceptance are precisely what branded consumer goods are made of.

- In a linear-demand/Bertrand example, Bresnahan derives the asymptotic bias. With national demand shocks, own-price coefficients are biased **toward zero**: demand looks steeper than it is.

- **The bias points in the direction of the finding.** To believe the estimate you must believe either (i) there are no national brand-level demand shocks, or (ii) they are unforecastable when prices are set. Neither is comfortable.

# Bresnahan's Critique II: Imperfect Competition

- Hausman computes General Mills' optimal re-pricing of Cheerios and Honey-Nut Cheerios **holding Kellogg's prices fixed**, and calls the result a lower bound on the welfare gain.
- Bresnahan: this is not a Bertrand equilibrium calculation. Bertrand says firms take rivals' *post-entry* prices as given — not that rivals' prices are unchanged by entry. Entry changes the game; the new equilibrium has to be re-solved.
- Economically, adding a product makes rivals' residual demands **flatter**, which typically *lowers* their prices. What Hausman computes is the price-discrimination problem of a multiproduct monopolist, not an equilibrium.
- So the "lower bound" claim is wrong: the omitted equilibrium response goes the *other* way and would raise the welfare gain.
- Note also the flip side Bresnahan draws out: a steep *dd* curve means large consumer surplus *and* large private rents. Demand estimates of this kind speak to the private return to innovation as much as the social return.

# Chaudhuri, Goldberg and Jia (2006): Setup

The mirror image of the new-goods problem: what is the welfare cost when products are **withdrawn**?

- Question: under the WTO TRIPS agreement India must enforce pharmaceutical product patents. What happens to prices and welfare?
- Setting: the fluoroquinolone (antibiotic) sub-segment of the Indian market, monthly product-level data, 1999–2000. Several molecules were still under US patent at the time.

- **Two-level AIDS**: upper level allocates expenditure across sub-segments of systemic anti-bacterials; lower level allocates within quinolones across "product groups".
- A product group is a **molecule** $\times$ **(domestic / foreign)** cell. Allowing domestic and foreign versions of the *same* molecule to be distinct products is the key modelling choice: it opens a **variety** channel, not just a price channel.

- Counterfactual: patent enforcement forces domestic products out. Prices of the survivors are re-optimized. This is exactly the Bresnahan complaint taken seriously.

# CGJ (2006): Virtual and Equilibrium Prices, Jointly

- Two sets of unknown prices, **simultaneously determined**: $p_i^V$, the virtual prices of withdrawn domestic products, and $p_j^*$, the new equilibrium prices of the survivors.
- The system is

$$\omega_i\left(p^V, p^*\right) = 0 \qquad \text{for withdrawn } i$$

$$p_j^* = mc_j - \left[\Omega \circ \left(\frac{\partial q}{\partial p}\right)\right]_j^{-1} q\left(p^V, p^*\right) \qquad \text{for surviving } j$$

  solved as one fixed point. Hausman's exercise is the special case in which the second block is frozen.

- Marginal costs are not observed. CGJ bracket them: an **upper** bound on $mc$ from single-product Bertrand, a **lower** bound from within-group joint profit maximization, then report a range.

- Plausibility check: compare simulated Indian multinational prices to **actual Pakistani prices** — similar demographics, but a market structure resembling the post-patent counterfactual.

# CGJ (2006): Decomposing the Welfare Loss

Three price vectors, hence three cleanly separated channels. Let $u^0$ be pre-TRIPS utility.

$P^1$: virtual prices for withdrawn products, *everything else fixed*, including quinolone expenditure.

**Pure variety effect** $= E(u^0, P^1) - E(u^0, P^0)$

$P^2$: as above, but quinolone *expenditure* adjusts to the new segment price index.

**Expenditure switching** $= E(u^0, P^2) - E(u^0, P^1)$

$P^*$: full equilibrium, surviving firms re-optimize.

**Reduced competition** $= E(u^0, P^*) - E(u^0, P^2)$

- $P^2$ is also the empirically relevant scenario if TRIPS is accompanied by **price regulation** that pins prices at pre-patent levels. Note that regulation removes the third channel but **not** the first two.
- This is a nice template: any counterfactual that removes products should report variety and price channels separately, because policy instruments act on them differently.

# CGJ (2006): Results

- Without price regulation, prices of foreign patented products rise by 100% **to** 400%.
- With price regulation (prices held at pre-TRIPS levels), the annual welfare loss from withdrawal of all four domestic product groups is about **Rs.** 13.7**bn** $\approx$ **US$**305**m** — roughly 50% of systemic anti-bacterial sales in 2000.
  - Foregone domestic producer profit is only about US$50m. **The bulk is consumer welfare.**
  - Foreign producers' profit gain is small: US$19.6m with regulation, US$53m without.
- Stripping out the variety component still leaves a price-driven loss worth about 24% of antibiotic sales — roughly a third of the upper bound.
- Losses are highly **non-linear** in how many domestic products exit: withdrawing one is modest (consumers switch to the others), withdrawing all four is severe — a direct consequence of the high estimated cross-elasticities *among* domestic products.
- Interpretation offered: the domestic/foreign distinction largely captures **ease of access** (distribution reach), suggesting compulsory licensing rather than price control — and possibly transitional.

# Turning to Characteristic Space

- What product space gave us: flexible substitution, an exact money metric, closed-form virtual prices.
- What it could not give us:
  - a way to price a good with *no* sales history at all (we needed the good to exist in order to fit its demand curve),
  - a tractable system when $J$ is in the hundreds without imposing a segment tree,
  - observed consumer heterogeneity.

- Characteristic space fixes all three by writing utility over **attributes** rather than names. A hypothetical product is then just a point in attribute space.

- The remainder of this lecture is a preview; Allan develops BLP in the next two lectures. The point here is to see **how welfare works** in this class of models, and how badly it can go wrong.

# Takeaways

1. **Demand estimation is the input to everything else** in empirical IO: markups, merger simulation, conduct tests, welfare.

2. **Dimensionality forces a choice**. Product space handles substitution flexibly but needs a separability tree; characteristic space scales but restricts substitution through the attribute space. Both assumptions are testable; test them.

3. **Welfare is an integral under a Hicksian curve.** *CS* is an approximation whose error Willig bounds; it is fine for a small good and a small price change, and not fine for the applications we care about.

4. **New (and disappearing) goods are just a price change** — to or from a virtual price. But that price is out of sample, so the answer is driven by functional form. Report bounds and sensitivity.

5. **Identification deserves as much attention as the model.** The Hausman IV is a genuinely clever solution to a hard problem and it may well be invalid in exactly the settings where it is most tempting.

6. **Counterfactuals need an equilibrium**, not a partial derivative. Let rivals re-optimize (CGJ), or say clearly that you did not.

# Readings: Core

- Deaton, A. and J. Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70(3), 312–326.
- Hausman, J., G. Leonard and J.D. Zona (1994), "Competitive Analysis with Differentiated Products," *Annales d'Economie et de Statistique* 34, 159–180.
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- Chaudhuri, S., P.K. Goldberg and P. Jia (2006), "Estimating the Effects of Global Patent Protection in Pharmaceuticals: A Case Study of Quinolones in India," *AER* 96(5), 1477–1514.

**Welfare background**

- Hausman, J. (1981), "Exact Consumer's Surplus and Deadweight Loss," *AER* 71(4), 662–676.
- Willig, R. (1976), "Consumer's Surplus Without Apology," *AER* 66(4), 589–597.